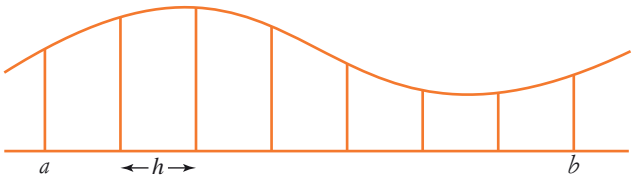
WORKSHEET

Trapezoidal rule

Area under a curve $\approx \int_a^b f(x)dx = \frac{1}{2}h\left[f(x_0) + f(x_n) + 2\sum_{i=1}^{n-1} f(x_i)\right]$ where the interval (a, b) is divided into n equal subintervals of length h; hence $h = \frac{b-a}{n}$.



1 Given the integral $\int_3^5 \frac{1}{x-1}dx$ with $n = 4$, find the value of:

a $f(x_0)$

b $f(x_1)$

c $f(x_2)$

d $f(x_3)$

e $f(x_n)$

2 Use the trapezoidal rule to approximate the area given by $\int_0^2 \frac{1}{x-1}dx$ with $n = 4$.

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3 Given the integral $\int_0^2 3^x \, dx$ with $n = 5$, find:

a $f(x_0)$

b $f(x_1)$

c $f(x_2)$

d $f(x_3)$

e $f(x_4)$

f $f(x_n)$

4 Use the trapezoidal rule to approximate the area given by $\int_0^2 3^x \, dx$ with $n = 5$.

5 Use the trapezoidal rule to approximate the area given by $\int_2^6 \frac{1}{x^2 + 1} \, dx$ with $n = 8$.

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6 A 100 m section of mountain is to be removed to make way for a road. Height measurements are taken at 20 m intervals and are recorded as 12 m, 14 m, 18 m, 19 m, 21 m, and 16 m. Use the trapezoidal rule to find the approximate area of the cross-section.

7 Geographers want to find the area of a cross-section of a river. The river is 40 m across and depth measurements of 3 m, 4 m, 6 m, 7 m, 6 m, 5 m, 6 m, 4 m, and 3 m are taken at 5 m intervals. Use the trapezoidal rule to approximate the area.

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Answers

1 **a** $\frac{1}{2}$

b $\frac{2}{5}$

c $\frac{1}{3}$

d $\frac{2}{7}$

e $\frac{1}{4}$

2 ≈ 0.6970 units^2

3 **a** 1

b ≈ 1.5518

c ≈ 2.4082

d ≈ 3.7372

e ≈ 5.7995

f 9

4 ≈ 7.3987 units^2

5 ≈ 0.3016 units^2

6 ≈ 1720 m^2

7 ≈ 205 m^2

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